

*ThermoQ*

# User Manual

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Version 1.0.1

A worked example on Al–Cu–Li (COST)

Using the calculation files shipped in test/

Switch the interface with Help → Language. Help → User Manual opens this PDF in the language currently selected. Windows users may install ThermoQ-1.0.1-Windows-Setup.exe from GitHub Releases (no Python required).

## 1. Scope

This manual follows one alloy, Al–Cu–Li, with the COST assessment, from Thermo-Calc and Pandat batch files through extraction and plotting. The figures are computed from the .exp / CSV files in test/; they are the same objects ThermoQ writes to Excel.

Menus: File, Import, Plot, Tools, Help. Calculate sits on the main window (single composition and composition-space batch). After Plot, figures preview in the window (Save As... / Open). Language labels refresh after Help → Language.

| Menu                                         | Purpose                                                                                      |
|----------------------------------------------|----------------------------------------------------------------------------------------------|
| Import → Pandat to ThermoQ                   | Load P.xlsx, Ts.xlsx (required); P-S.xlsx, Ts-S.xlsx (optional).                             |
| Tools → Composition Converter                | Convert wt% ↔ at% compositions.                                                              |
| Tools → Generate Thermo-calc Batch File      | Build a .tcm from Template0 + loop body (%Element%, %T%).                                    |
| Tools → Generate Thermo-Calc Batch Data File | Excel/CSV Data File for Property Model Calculator batch (General Models + Custom/Diffusion). |
| Tools → Extract Thermo-calc Results          | Melting Range, Miscibility Gap, T-zero, TriST Zone.                                          |
| Tools → Generate / Extract Pandat            | .pbfx batch (T-zero, Gibbs) and P/Ts, T0.xlsx, TriST.                                        |
| Plot                                         | Phase surfaces, Qtrue, liquidus vectors, k, T-zero, miscibility gap; in-window preview.      |
| Help → Language                              | Switch UI English / □□ (menus and open tool windows refresh).                                |
| Help → User Manual / Example                 | Open this PDF (by language) or the bundled Example/ folder.                                  |

### 1.1 Tools → Generate Thermo-Calc Batch Data File

Creates Property Model Calculator batch Data Files (.xlsx / .csv) with English headers (Id, element symbols, Composition unit, Temperature unit; optional Param ... / Exp ...). Choose a General Model (Coarsening, CET, Crack Susceptibility, Driving Force, Equilibrium, Freeze-in, Interfacial Energy, Liquidus/Solidus, Phase Transition, Scheil, Spinodal, T-Zero, Yield Strength) or Custom/Diffusion (Temperature column, as in Ti–Al–Mn–1173K.xlsx). One Balance (Bal) element plus a composition grid; the temperature grid maps to the model's Param (e.g. Evaluation / Annealing / Start temperature). Phase dropdowns stay on the Thermo-Calc GUI. Sample: Example/Generate Thermo-Calc Batch Data File/.

## 2. Installation

- Python 3.8+ with tkinter. `pip install -r requirements.txt` then `python main.py`.
- TriST workbooks require matplotlib and scipy. Interactive HTML requires plotly.

## 3. Case: melting range (Thermo-Calc)

Folder for templates: test/Generate Thermo-calc Batch File-Melting Range/Al-Cu-Li/. Folder for results: test/Extract Thermo-calc Results-Melting Range/Al-Cu-Li\_COST/exp/ (filenames such as Al0.050Cu0.010Li\_np-T.exp).

### 3.1 Generate the .tcm

1. Tools → Generate Thermo-calc Batch File.
2. Template0 = template0.txt (database, system, baseline s-c, first make file).
3. Loop template = template.txt. Placeholders %Cu% and %Li% are replaced on the Min/Max/Step grid (mass fraction).
4. Write the merged .tcm (Alltcm.tcm is already in that folder) and run it in Thermo-Calc. Each composition produces one \*\_np-T.exp.

#### Template0 (baseline composition, excerpt):

```
go da
sw tcal9
d-sys al cu li
l-s c
get
go p-3
s-c t=500,p=101325,n=1
s-c w(cu)=0.00001
s-c w(li)=0.00001
l-c
c-e
s-a-v 1 t 300 1000 12.5
save_workspace Al0.00Cu0.00Li_np-T.POL y
step normal
post
s-d-a x t
s-d-a y np(liquid)
make file Al0.00Cu0.00Li_np-T.exp
```

#### Loop body (one block per grid point):

```
back
s-c w(cu)=%Cu%
s-c w(li)=%Li%
l-c
c-e
s-a-v 1 t 300 1000 12.5
save_workspace Al%Cu%Cu%Li%Li_np-T.POL y
step normal
post
s-d-a x t
s-d-a y np(liquid)
make file Al%Cu%Cu%Li%Li_np-T.exp
```

### 3.2 Extract liquidus and solidus

1. Tools → Extract Thermo-calc Results → Melting Range. Select the exp folder. Default filter \*\_np-T.exp.
2. The filename supplies w(Cu) and w(Li). Blocks between “\$ PLOTTED COLUMNS ARE : T and NP(LIQUID)” and BLOCKEND are read.
3. Liquidus is the lowest T with NP(LIQUID) = 1; solidus is the highest T with NP(LIQUID) = 0. Optional Template1 writes a .tcm for files listed as errors in Status.

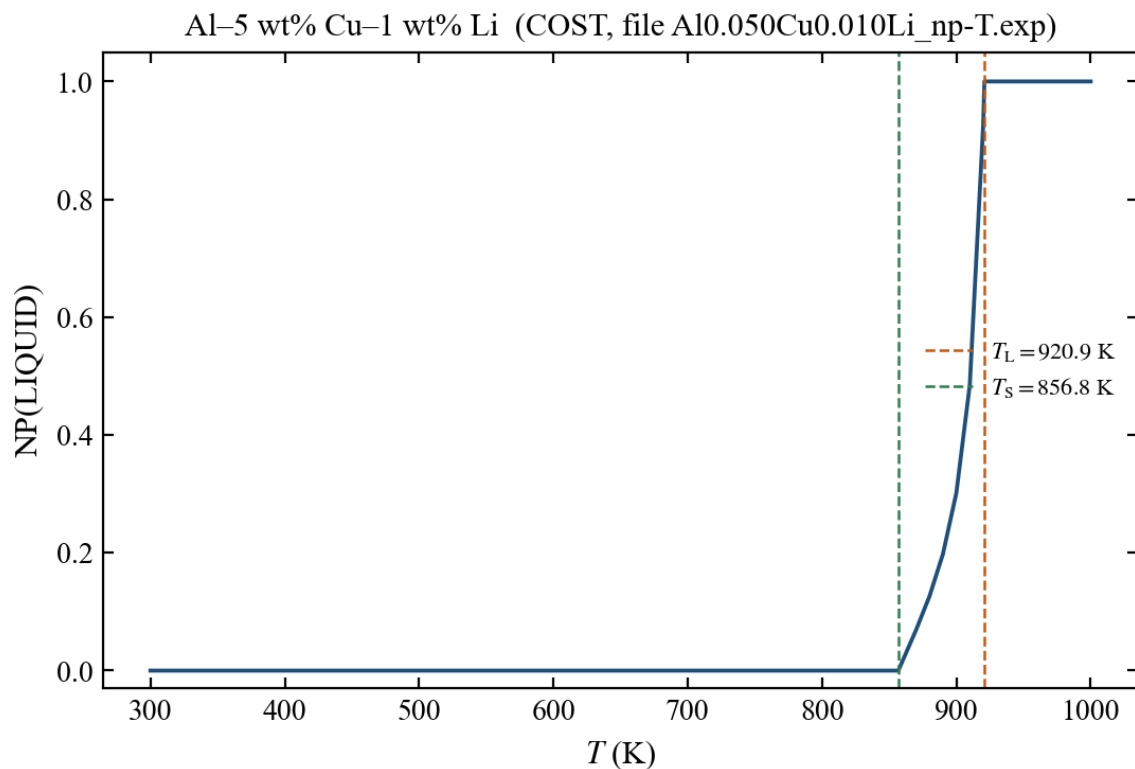


Fig. 1. NP(LIQUID) versus temperature for Al-0.050Cu-0.010Li (COST). Vertical lines are the liquidus and solidus reported in the extract Excel.

Five rows from that extract (same algorithm as the Melting Range tab):

| File                      | w(Cu) | w(Li) | TL / K | TS / K | $\Delta T$ / K |
|---------------------------|-------|-------|--------|--------|----------------|
| Al0.050Cu0.010Li_np-T.exp | 0.050 | 0.010 | 920.9  | 856.8  | 64.1           |
| Al0.100Cu0.020Li_np-T.exp | 0.100 | 0.020 | 904.2  | 843.2  | 61.0           |
| Al0.150Cu0.030Li_np-T.exp | 0.150 | 0.030 | 880.7  | 836.8  | 43.9           |
| Al0.180Cu0.005Li_np-T.exp | 0.180 | 0.005 | 883.1  | 814.6  | 68.5           |

Extract Thermo-calc Results → Melting Range · Al-Cu-Li COST

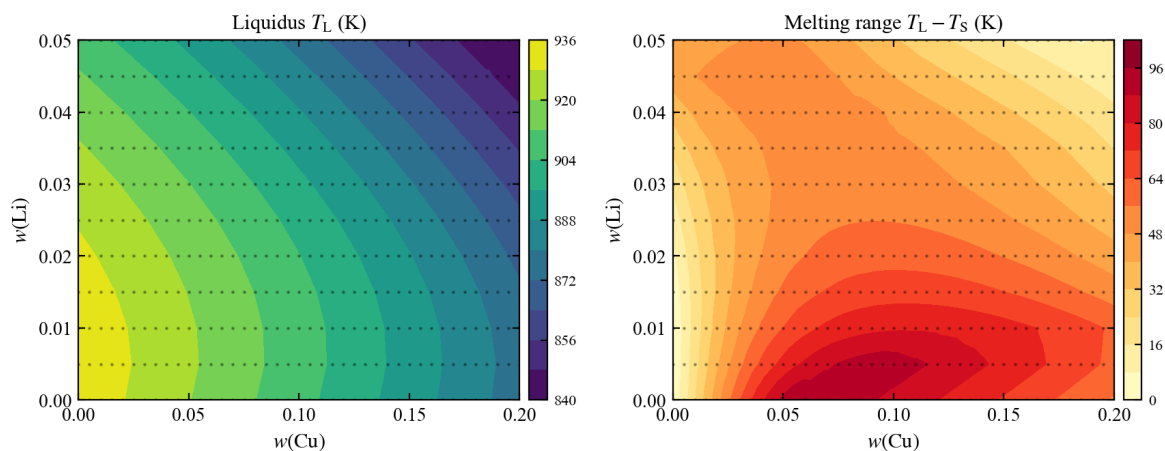


Fig. 2. Liquidus and melting range over the Al-Cu-Li COST grid in test/.../Al-Cu-Li\_COST/exp. Dots are individual .exp files.  
Plot → Phase Surfaces → Thermo-calc loads the Excel of this map.

## 4. Case: T-zero

Templates: test/Generate Thermo-calc Batch File-T0/Al-Cu-Li/. Results: test/Extract Thermo-calc Results-T0/Al-Cu-Li\_COST/exp/AlCu-0.030Li\_T0.exp, etc. The loop holds  $w(\text{Li})$  fixed and scans  $w(\text{Cu})$  from 0 to 0.2.

```
back
s-c w(Li)=%LI%
l-c
c-e
S-A-V 1
w(Cu)
0
0.2
0.005
save AlCu-%LI%Li_T0.POL y
step
t-z
FCC_A1
liquid
po
s-d-a x w(Cu)
s-d-a y t-k
make file AlCu-%LI%Li_T0.exp
```

1. Extract Thermo-calc Results → T-zero. Filename →  $w(\text{Li})$ ; XTEXT  $W(\text{Cu})$  → scan axis; plotted Y →  $T_0$  (K).
2. Save `t_zero.xlsx`. Plot → Plot T-zero Surface (Thermo-Calc), set the isotherm interval.

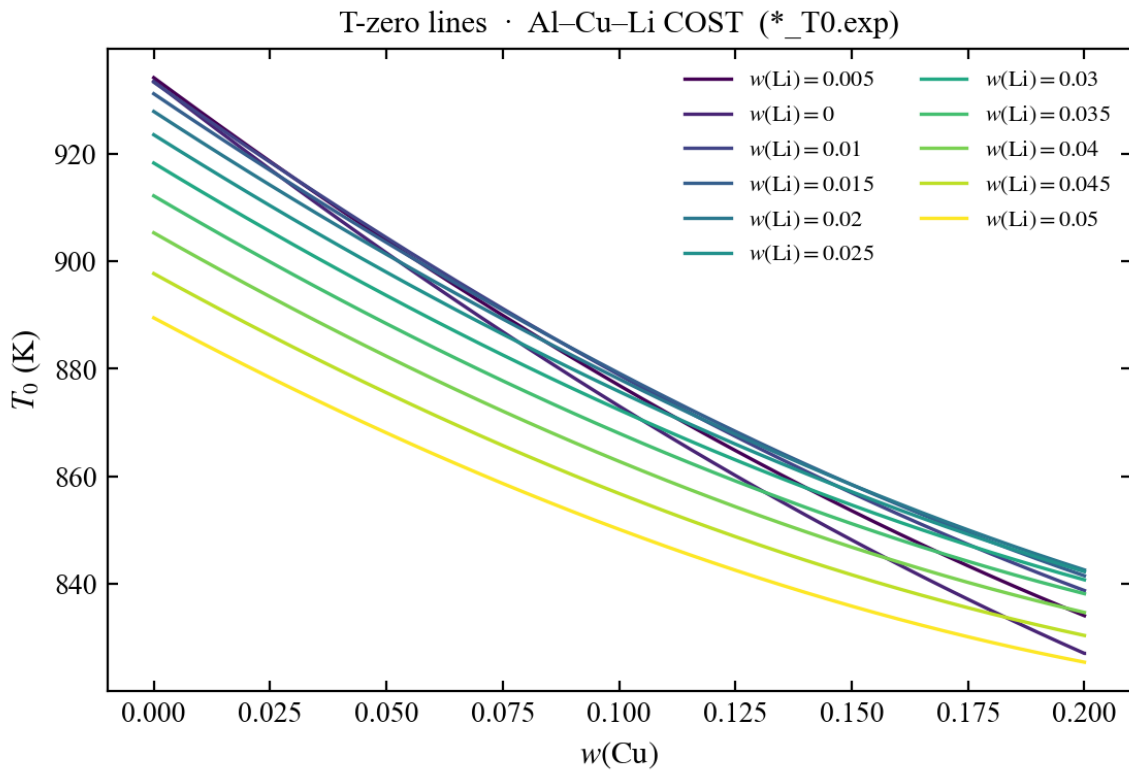


Fig. 3.  $T_0(w(\text{Cu}))$  at fixed  $w(\text{Li})$  from every \*\_T0.exp in the COST folder.

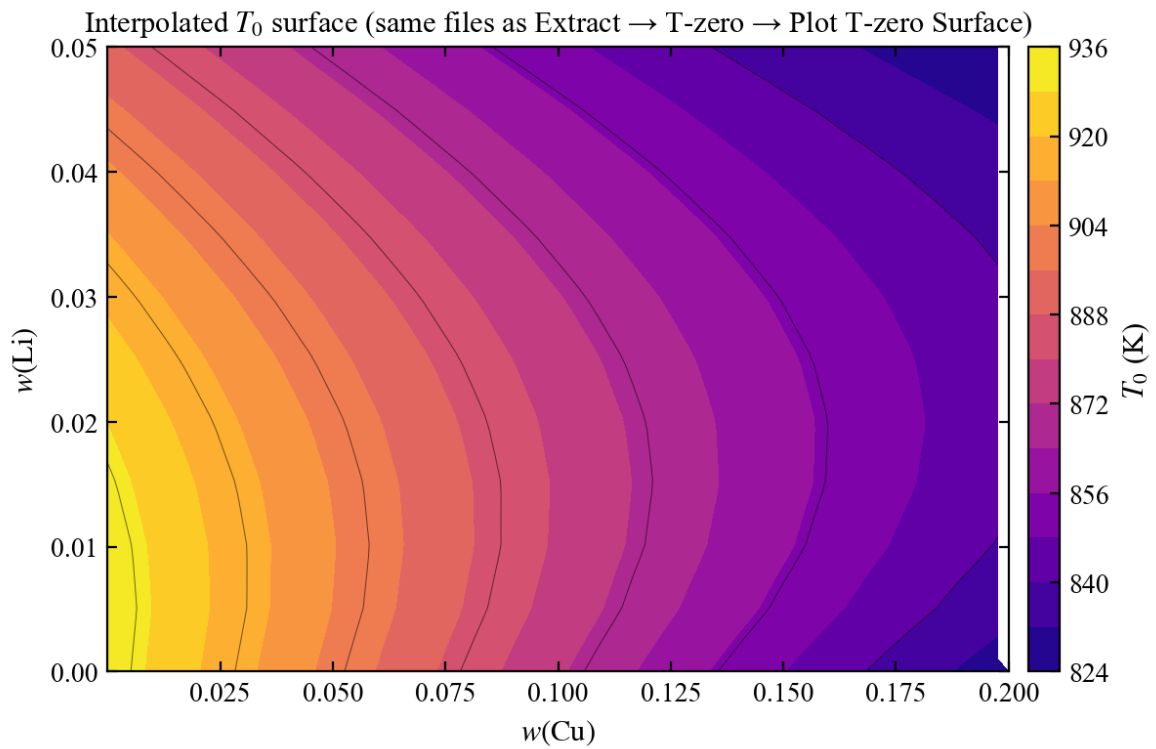


Fig. 4. Linear interpolation of those lines on the  $w(\text{Cu})$ – $w(\text{Li})$  plane — the field drawn by Plot T-zero Surface.

## 5. Case: liquid miscibility gap

Files: test/Extract Thermo-calc Results-Miscibility Gap/Liquid/Al-Cu-Li\_COST/exp/AlCuLi\_800.exp (one temperature per file). Extract → Miscibility Gap parses T from the name, and XTEXT/YTEXT as mole % Cu and mole % Li.

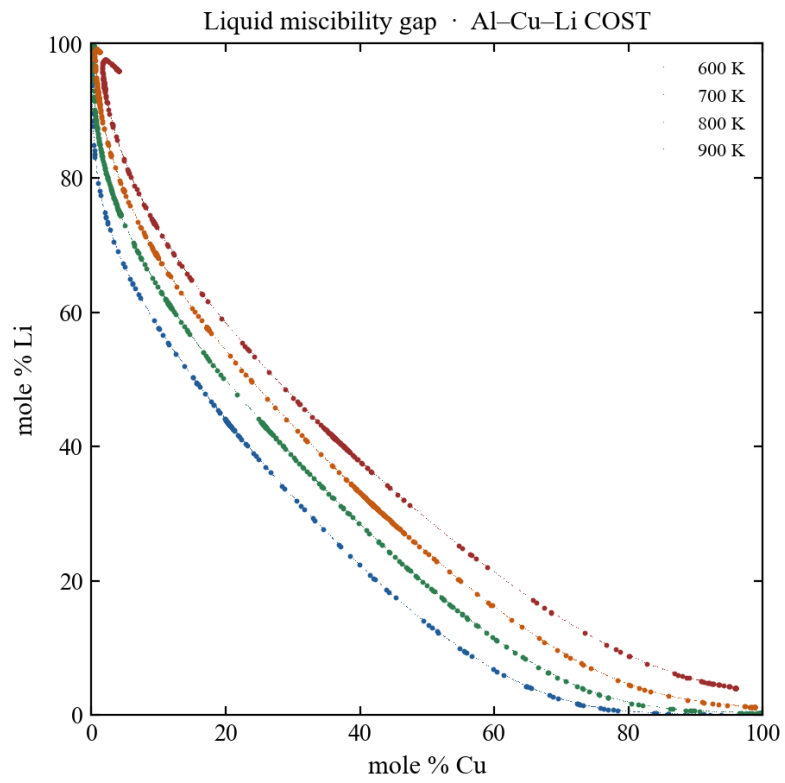


Fig. 5. Liquid miscibility-gap traces at 600–900 K from the COST .exp set. Plot → Plot Miscibility Gap reconstructs the same boundaries (with interpolation at a requested T).

## 6. Case: Gibbs energies and TriST Zone

Files: test/Extract Thermo-calc Results-Gibbs/Al-Cu-Li\_COST/exp/, e.g. AlCu-0.050Li\_650T\_Gibbs.exp. Each file is one (w(Li), T); the scan axis is w(Cu). Sections are labelled “\$ PLOTTED COLUMNS ARE : W(CU) and GMR(PHASE)”.

1. Extract Thermo-calc Results → TriST Zone. Folder search is recursive; default filter `*_Gibbs\exp$`. Axes X/Y fill after Browse.
2. Build TriST workbook (T0\_tie\_1D, T0\_lines, TriST\_boundaries, TriST\_mask, optional \_trist\_cube.npz). Grid N is typically 40–80.
3. Failed files go to Status; Template1 can emit one .tcm per error. Visualize with Plotly / 2D / 3D / GIF.

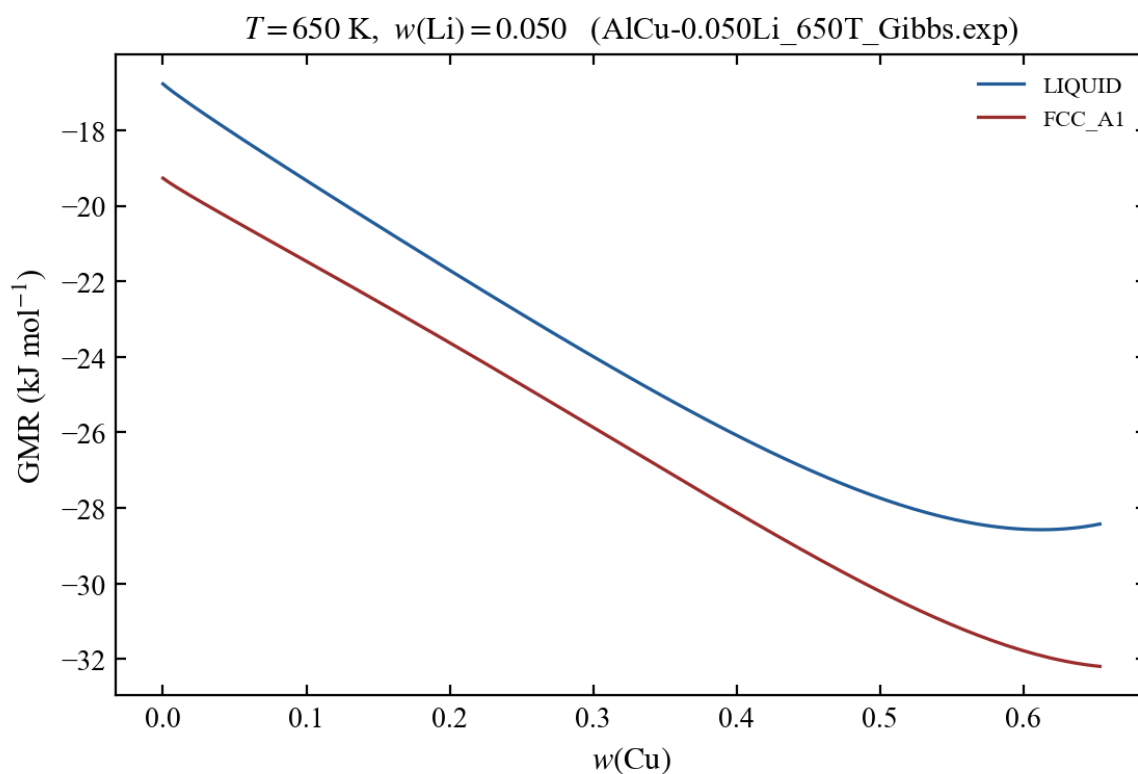


Fig. 6. Molar Gibbs energy of LIQUID and FCC\_A1 versus w(Cu) at 650 K and w(Li) = 0.050, taken from one COST Gibbs .exp.



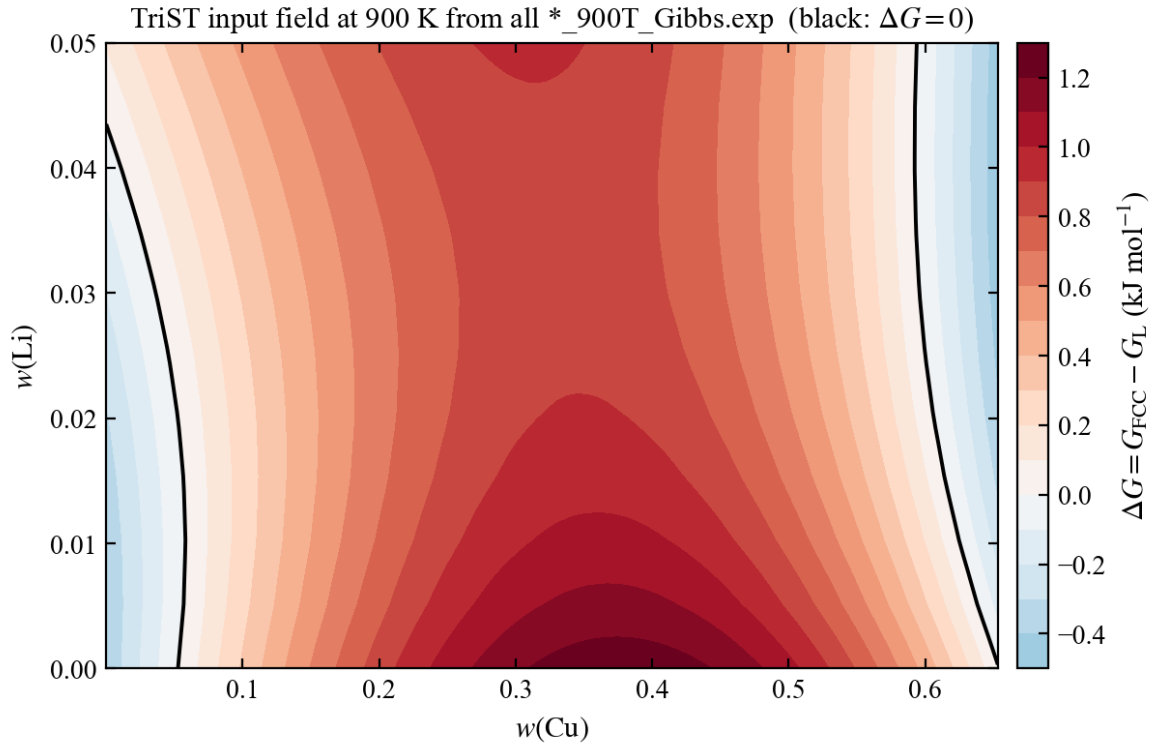


Fig. 7.  $\Delta G = G(\text{FCC}) - G(\text{L})$  at 900 K assembled from all \*\_900T\_Gibbs.exp in the same folder. The black contour is  $\Delta G = 0$ , i.e. the  $T_0$  / TriST boundary at that temperature.

## 7. Case: Pandat Lever table (same alloy)

Batch templates: test/Generate Pandat Batch File-T0/Al-Cu-Li\_COST/ and ...-Gibbs/. Solidification tables: test/Extract Pandat Results-P Ts (Lever Scheil)/Al-Cu-Li\_COST/. Extract Pandat Results → P/Ts writes P.xlsx / Ts.xlsx / P-S.xlsx / Ts-S.xlsx, then Import to ThermoQ. Calculate uses Newton interpolation on those tables for  $Q_{\text{true}}$ ,  $Q/P/\beta$ ,  $\Delta T$  and  $\Delta T_s$ .

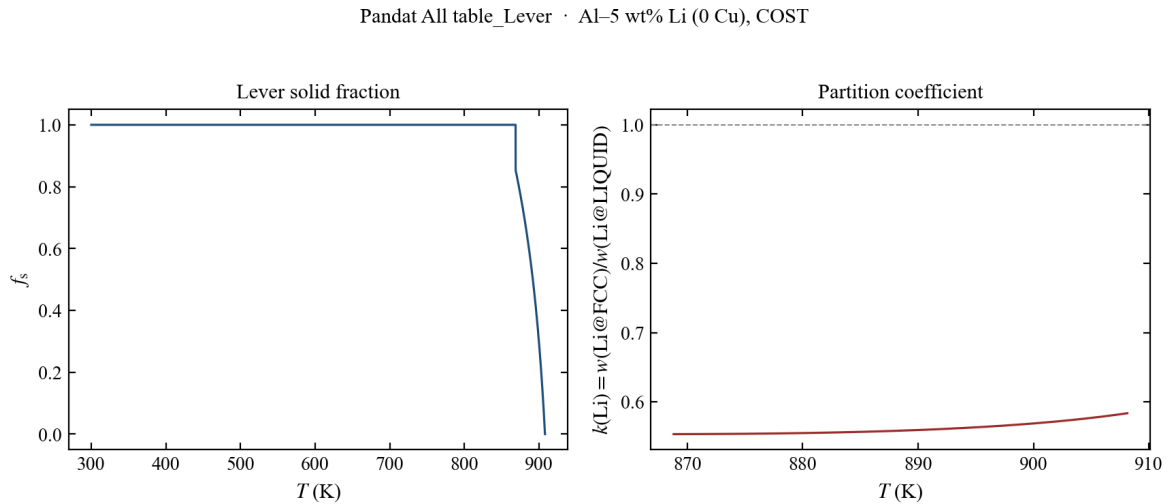


Fig. 8. One Lever CSV from that folder (Al-5 wt% Li, 0 Cu): solid fraction and  $k(\text{Li})$ . Plot → Solid-Liquid Partition Coefficients uses the same  $k = w(*@solid)/w(*@LIQUID)$ .

## 8. Plot tools after extract / import

- Phase Surfaces — Pandat P/Ts or the melting-range Excel of §3.
- Qtrue, liquidus vectors, partition k (liquidus / isotherm / isocomposition).
- T-zero Surface and Miscibility Gap — Excel from §4 and §5.
- Plot Labels: empty fields keep default axis names.

## 9. Where the other test/ trees fit

| test/ folder                                                                 | ThermoQ command                 |
|------------------------------------------------------------------------------|---------------------------------|
| Generate Thermo-calc Batch File-{Melting Range, T0, Gibbs, Miscibility Gap}/ | Generate Thermo-calc Batch File |
| Extract Thermo-calc Results-Melting Range/                                   | Extract → Melting Range         |
| Extract Thermo-calc Results-T0/                                              | Extract → T-zero                |
| Extract Thermo-calc Results-Gibbs/                                           | Extract → TriST Zone            |
| Extract Thermo-calc Results-Miscibility Gap/                                 | Extract → Miscibility Gap       |
| Generate Pandat Batch File-T0 / -Gibbs/                                      | Generate Pandat Batch File      |
| Extract Pandat Results-P Ts (Lever Scheil)/                                  | Extract Pandat → P/Ts           |
| Extract Pandat Results-T0/                                                   | Extract Pandat → T-zero         |
| Extract Pandat Results-Gibbs/                                                | Extract Pandat → TriST Zone     |

The same layout is repeated for Al–Cu–Si, Al–Mg–Li, Al–Mg–Si and Ti–Fe–Cu, with COST, TCAL9, He2009, Peisheng2011 or Bo2013 in the folder name. Help → Example opens the smaller Example/ tree (includes Generate Thermo-Calc Batch Data File/Ti-Al-Mn-1173K.xlsx).

## 10. Notes

- Close an extract workbook in Excel before overwriting it.
- Abnormal-point TCM files require a Template1 with %Element% and %T%, after Status has recorded failures.
- TriST needs scipy and matplotlib.